



COMPUTER SECURITY

2025

FINAL 2023

(Solved)



SENIOR
25

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Choose the correct answer:

1.	Which type of firewall intercepts all communication between an internal network and the internet?							
	a)	Packet-filtering firewall	b)	Application firewall	c)	Stateful inspection firewall	d)	Proxy firewall
2.	Which type of firewall can be used to control access to packets on the basis of packet address?							
	a)	Packet-filtering firewall	b)	Application firewall	c)	Stateful inspection firewall	d)	Proxy firewall
3.	What is the <u>primary function of an IDS?</u>							
	a)	To prevent all unauthorized access to a network						
	b)	To monitor network traffic to prevent potential security breaches						
	c)	To encrypt all network traffic to prevent eavesdropping						
	d)	To block all incoming network traffic except for authorized sources						
4.	What is the difference between a network-based IDS and host-based IDS?							
	a)	A network-based IDS monitors network traffic while a host-based IDS monitors system logs						
	b)	A network-based IDS monitors system logs while a host-based IDS monitors network traffic						
	c)	A network-based IDS is installed on individual hosts while a host-based IDS monitors network traffic						
	d)	A network-based IDS is installed on the network while a host-based IDS is installed on individual hosts						
5.	Which type of IDS analyzes <u>network traffic</u> in real-time and can respond to threats immediately?							
	a)	Signature-Based IDS	b)	Anomaly-based IDS	c)	Host-based IDS	d)	Network-based IDS
6.	Which type of IDS monitors for <u>deviations</u> from normal network behaviour and can detect <u>previously unknown threats</u> ?							
	a)	Signature-Based IDS	b)	Anomaly-based IDS	c)	Host-based IDS	d)	Network-based IDS
7.	What is the raw transposition cipher ?							
	a)	A substitution cipher that rearranges the order of letters in a message						
	b)	A cipher that replaces each letter in a message with a numerical value						
	c)	A cipher that applies a mathematical formula to each character in a message						
	d)	A cipher that rearranges the order of words in a message						
8.	Which of the following fully describes session hijacking attack ?							
	a)	Intercepting and carrying on a session begun by another entity						
	b)	A malicious admin is monitoring all sessions and passes files to enemy						
	c)	Injecting something into the communication						
	d)	One host pretends to be another						
9.	<u>When a host pretends to be another</u> , for example: using his IP's, is known as _____ attack							
	a)	Masquerade	b)	Impersonation	c)	Session hijacking	d)	Man-in-the middle
10.	Denning's model is an example for _____							
	a)	Signature based IDS	b)	Heuristic based IDS	c)	Virus scan software	d)	Firewall Model



11.	Which profile is recommended for auditing?				
	a)	Logon and session activity	b)	Command or program usage	c) File access activity d) All are true
12.	A company wants to set up an online price list so that outsiders can see the products and prices offered, it wants to be sure that <u>no outsider can change the prices or product list</u> and <u>the outsider can access only the price list, not any of the more sensitive files stored inside</u> , the recommended firewall type will be _____				
	a)	Proxy gateway	b)	Application proxies	c) Guards d) Personal firewalls
13.	A university wants to allow its students to use e-mail up to a limit of so many messages or so many characters of e-mail in the last so many days . Although this result could be achieved by modifying e-mail handlers, it is more easily done by monitoring the common point through which all e-mail flows, the mail transfer protocol. The recommended firewall type will be _____				
	a)	Proxy gateway	b)	Application proxies	c) Guards d) Personal firewalls
14.	In context of OS security, the owner can protect an object by declaring it to be public or private , this concept is known as:				
	a)	Share via access limitation	b)	Share all or share nothing	c) Don't protect mode d) Isolate mode
15.	When OS runs different processes of different security requirements on the same time with shared resources that no other processes exist _____				
	a)	Cryptographic separation	b)	Logical separation	c) Temporal separation d) Physical separation
16.	<u>Physical OS separation</u> approach _____ the resource sharing and _____ the security level				
	a)	Minimizes , minimizes	b)	Minimizes , maximizes	c) Maximizes , minimize d) Maximizes, Maximizes
17.	If different processes can run concurrently unaware of the presence of each other , and each process has its own address space, files, and other objects, this OS level of sharing is categorized as _____				
	a)	Isolation level	b)	Share via access limitations	c) Share via capabilities d) Share all or share nothing
18.	The mechanisms used to confirm a user's identity is known as:				
	a)	Identification	b)	Authentication	c) Authorization d) Access Limitation
19.	A kind of malicious software runs when a computer is started , it changes control on the OS instructions				
	a)	Memory resident virus	b)	Boot sector virus	c) One time execution d) Appended virus
20.	What is true about program flaws?				
	a)	It could be validation error or domain error	b)	It could be malicious or non-malicious	c) It could lead to failure d) All are true
21.	A type of virus which is active as long as the computer running				
	a)	Memory resident virus	b)	Brain virus	c) Polymorphic virus d) Logic bomb virus
22.	Controls against program threats include the following actions except _____				
	a)	Peer review	b)	Independent testing	c) Proof of correctness d) Decapsulation of different process



23.	Each virus has a signature that could be used to detect its existence, such as:							
	a)	Location	b)	Execution performance	c)	Spreading	d)	All are true
24.	Viruses resident in <u>application programs and libraries</u> are categorized as :							
	a)	One time execution virus	b)	Boot sector virus	c)	Other homes of virus	d)	Memory resident virus
25.	A program virus attaches itself to a program; whenever the program is run, the virus is activated . This kind of attachment is usually known as _____							
	a)	Integrated virus	b)	Surrounded virus	c)	Appended virus	d)	Replacement virus
26.	Example of one-way hash functions are:							
	a)	MD4, MD5, SHA-512	b)	DES, 2DES	c)	AES, 3DES	d)	Diffie-Hellman
27.	Digital signatures provide the ability to:							
	a)	Verify author, data & time of signature	b)	Authenticate message contents	c)	Be verified by third parties to resolve disputes	d)	All are true
28.	All of the following algorithms <u>could be used for key session exchange</u> except :							
	a)	Diffie-Hellman	b)	Elliptic curve	c)	DSS	d)	RSA
29.	What is FALSE about Kerberos authentication protocol?							
	a)	It is a two-party one-way authentication protocol						
	b)	It requires two special function servers						
	c)	It is always a third party						
	d)	It used symmetric key encryption						
30.	The ticket issued to the user by Kerberos protocol includes the following data							
	a)	User login details	b)	User privilege on the accessed application	c)	Time-date stamp	d)	All are true
31.	What is true about Steganography?							
	a)	It is used to encrypt images and send it in unreadable form						
	b)	It uses images to hide important data in LSB of each image's pixel						
	c)	The encrypted file has smaller size compared with the original one						
	d)	It can be broken by brute-force attack						
32.	Example for classical substitution encryption							
	a)	Playfair encryption	b)	DES	c)	Block or row transposition classical techniques	d)	AES
33.	Triple-DES with two keys is defined by $C=EK_3(DK_2(EK_1(P)))$, if K1 equals K3 , the triple DES can be considered							
	a)	2DES	b)	3DES	c)	DES	d)	Useless encryption



34.	Which of the following is a good description for Triple-DES ?							
	a)	C =EK1(DK2(EK1(P)))	b)	C =EK2(DK1(EK1(P)))	c)	C =EK3(EK2(EK1(P)))	d)	C = DK3(DK2(DK1(P)))
35.	_____ allow key exchange without real-time access to public-key authority							
	a)	Certificate authority	b)	Public key authority	c)	Public announcement	d)	Public direction
36.	_____ is a type of attack that depends on analyzing relations among the contents on cipher text							
	a)	Time attack	b)	Statistical analysis	c)	Dictionary attack	d)	Brute force
37.	Erasure of program or data or malfunction of an OS file manager is considered							
	a)	Modification	b)	Interception	c)	Interruption	d)	Fabrication
38.	Cyber-security rules for daily practicing are classified as							
	a)	HW control	b)	SW Control	c)	Administration control	d)	All
39.	It means that some unauthorized party (person, program) has gained access to data							
	a)	Interception	b)	Modification	c)	Interruption	d)	Fabrication
40.	Legal and ethical controls are classified as							
	a)	Hardware controls	b)	Software controls	c)	Policy controls	d)	Physical controls
41.	In Caesar cipher, <u>shifting a letter by +3 positions</u> means that:							
	a)	A becomes D, B becomes E, C becomes F, and so on						
	b)	A becomes X, B becomes Y, C becomes Z, and so on						
	c)	A becomes C, B becomes F, C becomes I, and so on						
	d)	A becomes B, B becomes C, C becomes D, and so on						
42.	What is a weakness of Caesar cipher?							
	a)	Vulnerable to brute-force attacks						
	b)	Vulnerable to frequency analysis						
	c)	Both a & b						
	d)	Not vulnerable to any type of attacks						
43.	What is the number of rounds in DES?							
	a)	8	b)	12	c)	16	d)	24
44.	What is the weakness of DES?							
	a)	Vulnerable to brute-force attacks	b)	Vulnerable to differential cryptanalysis	c)	Both a&b	d)	Not Vulnerable to any type of attack
45.	Which of the following is a common technique used with transposition cipher ?							
	a)	Rail fence cipher	b)	Caesar cipher	c)	Vigenère cipher	d)	Playfair cipher



46.	What is a disadvantage of rail fence cipher compared to other transposition cipher?							
	a)	It is more complex to implement	b)	It has a smaller key space	c)	It's slower in encrypting and decrypting	d)	It is more vulnerable to attacks
47.	What is the block size of <u>triple DES</u> ?							
	a)	64-bit	b)	128-bit	c)	256-bit	d)	512-bit
48.	What is a common example of a man in the middle attack?							
	a)	Phishing	b)	Denial-of-service attack	c)	SQL injection	d)	Wi-Fi eavesdropping
49.	How can a man-in the middle attack be prevented?							
	a)	Using strong encryption	b)	Using secure authentication methods	c)	Using digital signature	d)	All of the above
50.	How can users protect themselves from a phishing attack?							
	a)	Being cautious of messages or websites that ask for sensitive information						
	b)	Checking the URL of a website before entering any information						
	c)	Keeping software and antivirus programs up-to-date						
	d)	All of the above						